FINAL PROJECT REVIEW | ACADEMIC PRESENTATION

AI-Based Structural Variant Prioritization from Long-Read Sequencing Using Explainable Machine Learning

A Deep Learning & Multimodal Framework with Conformal Risk Guarantees and TreeSHAP Explainability for Clinical Diagnostics

PROJECT GROUP

Group 15
B.Tech Final Year

DEPARTMENT

Department of Computer Science
and Engineering

DATA MATRIX

ClinVar (101k SVs)
gnomAD SV v2.1 (387k)

VALIDATION

Chr 19-22, X, Y Holdout
0.0% Locus Overlap

BACKGROUND

Introduction: Third-Generation Long-Read Sequencing

Long-read sequencing resolves dark genomic regions but introduces vast interpretation challenges

Long-Read Sequencing Revolution

PacBio HiFi & Oxford Nanopore

- Single-molecule long reads (>10,000–25,000 bp) span repetitive genomic regions.
- Overcomes short-read mapping blindness in tandem repeats and segmental duplications.
- Routinely uncovers >25,000 structural variants (>= 50 bp) per individual genome.

Structural Variant Impact

Pathogenicity & Disease Burden

- SVs alter more total base pairs than all single-nucleotide variants (SNVs) combined.
- Primary drivers of rare congenital diseases, autism, developmental delay, and cancer.
- Include complex deletions, duplications, inversions, translocations, and insertions.

The Clinical Triage Bottleneck

Automated Scoring Deficit

- Overwhelming volume of candidate SVs in clinical long-read VCF files.
- Existing clinical tools focus almost exclusively on SNVs (CADD, REVEL, SIFT).
- Urgent need for automated, explainable machine learning prioritization algorithms.

PROBLEM FORMULATION

Problem Statement: High-Dimensional Variant Ambiguity

Critical computational gaps preventing routine clinical adoption of long-read sequencing

01

Tool & Method Inadequacy

Over 90% of current pathogenicity scoring tools are engineered exclusively for SNVs or micro-indels (<50 bp). They fail to model multi-kilobase structural context, exon-spanning disruptions, and topological chromatin domain boundaries.

02

Genomic Data Leakage

Prior studies evaluate SV prioritization using naive random k-fold train/test splits. Overlapping variants from identical genomic loci or genes leak between folds, producing artificially inflated benchmark metrics that fail on real patients.

03

Lack of Certified Safety

Conventional AI models output uncalibrated probability scores without uncertainty quantification. In high-stakes clinical diagnostics, arbitrary cutoffs risk missing pathogenic mutations or triggering costly unnecessary medical interventions.

CLINICAL RELEVANCE

Clinical Motivation: Resolving the 40% VUS Dilemma

Transforming non-actionable genomic findings into certified diagnostic intelligence

~40%

Variants of Uncertain Significance
Up to 40% of discovered patient SVs cannot be clinically classified.

>45 min

Manual Review per Variant
Exhaustive cytogenetic and PubMed literature searches stall reporting.

4.2s

Automated AI Pipeline Delivery
Our proposed workflow ranks and explains candidate variants in seconds.

CLINICAL IMPLICATIONS & IMPACT

Why Solving VUS Prioritization is Paramount

Delayed Medical Interventions: When a candidate mutation is tagged as VUS, oncologists and medical geneticists cannot prescribe targeted therapy or gene-editing treatments.

Severe Diagnostic Backlog: Clinical geneticists spend hours cross-referencing population frequencies, constraint metrics, and gene catalogs for hundreds of variants per patient.

Explainability Requirement: Clinicians cannot trust black-box neural scores. Automated prioritization must supply local biological feature attributions for hospital board validation.

STATE OF THE ART

Literature Review: Genomic Variant Interpretation

Foundational scientific milestones and existing scoring methodologies

Study & Year	Methodology	Key Findings & Strengths	Inherent Limitations
Collins et al. (Nature 2020)	gnomAD SV v2.1 Population Catalog	Mapped 387,700 SVs across 14,891 genomes; established allele frequency baseline.	Purely reference catalog; lacks disease pathogenicity predictive scoring.
Rentzsch et al. (Genome Bio 2020)	CADD-SV Regional Scoring	Integrates functional genomics and conservation tracks for large deletions.	Linear kernel; struggles with non-coding complex balanced rearrangements.
Landrum et al. (NAR 2024)	NCBI ClinVar Database	Gold-standard repository of 100k+ expert-asserted variant-disease records.	Suffers from heavy class imbalance and unclassified VUS backlog.
Lundberg et al. (Nat MI 2020)	TreeSHAP Attribution	Game-theoretic local Shapley attributions for complex tree ensembles.	Diagnostic tool only; requires integration into end-to-end clinical pipelines.

RESEARCH MOTIVATION

Research Gap: Critical Unaddressed Challenges

Three major limitations in current genomic machine learning frameworks

GAP 1: EVALUATION INTEGRITY

Genomic Locus Leakage in Cross-Validation

Most published genomic AI models evaluate performance using standard random splits. Because structural variants frequently cluster around recombination hotspots and shared genes, variants from the same locus appear in both training and test sets. This causes severe data leakage and artificially inflated performance estimates.

GAP 2: REPRESENTATION RIGIDITY

Tree Step-Function Discretization

Popular gradient boosted trees (XGBoost, LightGBM) split continuous features using axis-aligned orthogonal boundaries. However, population allele frequencies follow continuous exponential decays under purifying selection. Decision trees fail to smoothly represent these manifolds, resulting in elevated false negative rates.

GAP 3: CLINICAL UNCERTAINTY

Lack of Certified Risk Guarantees

Existing algorithms output point probability estimates and apply subjective classification thresholds (e.g. tau = 0.50). In clinical genetics, subjective thresholds fail to distinguish between well-calibrated confidence and epistemic uncertainty, lacking distribution-free mathematical coverage guarantees.

PROJECT SCOPE

Aim, Objectives and Contributions

Formal project scope, methodology objectives, and achieved scientific contributions

PRIMARY AIM & OBJECTIVES

Engineering Objectives

- Harmonize multi-source genomic databases (ClinVar, gnomAD SV, 1000G) via 50% reciprocal overlap.
- Formulate a 24-dimensional non-leaking feature vector with strict chromosomal disjoint holdouts.
- Architect Deep Tabular ResNet and Multimodal Sequence-Epigenetics neural networks.
- Deploy Split Conformal Prediction (95% certified coverage) and TreeSHAP explainability in a production CLI tool.

KEY SCIENTIFIC CONTRIBUTIONS

Demonstrated Outcomes

- Eliminated Genomic Leakage: Enforced 0.0% locus overlap via Chr 19–22, X, Y disjoint testing.
- Benchmark Superiority: Peak 0.9705 PR-AUC and 94.4% clinical sensitivity on unseen chromosomes.
- Drastic False Negative Reduction: Slashed missed pathogenic variants from 1,745 down to 414 (>60% reduction).
- Clinical Backlog Triage: Resolved 48.8% of 34,466 ClinVar VUS into actionable risk tiers.

SYSTEM ARCHITECTURE

Proposed System Architecture

Modular end-to-end framework translating patient sequencing data into actionable clinical intelligence

01

Data Ingestion

Multi-Source Integration

- Parse Sniffles2/cuteSV long-read VCFs
- Filter high-confidence SVs (>= 50 bp)
- 50% reciprocal coordinate overlap
- Ingest ClinVar & gnomAD SV catalogs

02

Feature Vector

Zero-Leakage Engineering

- Compute 24 biological features
- gnomAD global AF & 7 continental AFs
- pLI haploinsufficiency & LOEUF scores
- Chromosomal disjoint holdout split

03

Dual-Paradigm AI

DL & Multimodal Inference

- PyTorch Tabular ResNet (skip connects)
- Multimodal Net (DNA seq + 5mC)
- Outputs continuous calibrated index
- 94.4% clinical sensitivity

04

Clinical Safety

Certified Explainability

- Split Conformal Prediction (95% guarantee)
- Local TreeSHAP waterfall attributions
- Production CLI tool (prioritize_sv.py)
- Interactive HTML & triage queue export

GENOMIC REPOSITORIES

Dataset Description and Sources

Multi-cohort genomic databases underpinning model development and independent validation

Source Database	Total Variants	Genomic Scope	Role in Pipeline
NCBI ClinVar (2024)	**101,232 SVs**	Deletions, Duplications, Inversions, Insertions	Clinical ground-truth labels (Pathogenic vs Benign)
gnomAD SV v2.1	**387,700 SVs**	14,891 genomes across 7 continental populations	Baseline allele frequency spectrum and PopMax
1000 Genomes (Phase 3)	**68,818 SVs**	2,504 multi-ethnic individuals	Independent cross-validation for polymorphic SVs
Final Harmonized Matrix	**44,268 SVs**	High-confidence labeled cohort (>= 50 bp)	**35,414 Train / 8,854 Chromosomal Holdout Test**

DATA ENGINEERING

Data Preprocessing and Harmonization

Standardizing heterogeneous multi-database coordinates and resolving interval overlaps

Coordinate Standardization

GRCh38 Reference Genome

All coordinates were mapped to the GRCh38/hg38 human reference assembly. Chromosomal nomenclature was unified to strict numerical and sex chromosomes (chr1–chr22, chrX, chrY). Unplaced contigs and centromeric gaps were pruned.

50% Reciprocal Overlap

Precise Interval Matching

Variants between patient VCFs and gnomAD are matched strictly using reciprocal coordinate overlap: (A ∩ B) / size(A) >= 0.50 and (A ∩ B) / size(B) >= 0.50. Prevents spurious cross-matches between adjacent distinct variants.

Label Consolidation

Binary Ground Truth Matrix

ClinVar assertions were mapped to binary ground truth: Pathogenic & Likely Pathogenic (Y = 1); Benign & Likely Benign (Y = 0). Ambiguous VUS records (N = 34,466) were segregated for post-training downstream clinical triage.

GENOMIC ANALYSIS

Exploratory Genomic Data Analysis

Strong biological divergence between pathogenic and benign structural variants confirms ML signal

Variant Type Dominance

Deletions vs Duplications

58.1%

Deletions in Pathogenic SVs

Deletions dominate pathogenic assertions because loss of coding exons causes irreversible haploinsufficiency. Duplications (40.7%) are more frequently tolerated as copy-number polymorphisms unless overlapping dosage-sensitive genes.

Bimodal Size Distribution

Retrotransposon Hotspots

~300 bp & ~6 kb

Distinct Structural Peaks

Structural variants follow a pronounced bimodal length spectrum. The first peak (~300 bp) aligns with Alu element retrotransposition; the second peak (~6,000 bp) corresponds to LINE-1 (L1) mobile element insertions and NHEJ double-strand break repair.

Purifying Selection Signal

Extreme Frequency Shift

p < 1e-300

Mann-Whitney U Test

Pathogenic structural variants undergo intense negative purifying selection in human populations (mean gnomAD AF = 0.00012). In contrast, benign variants reach high allele frequencies (mean AF = 0.0418). Proves popgen frequency is a top diagnostic predictor.

METHODOLOGICAL RIGOR

Feature Engineering and Data Leakage Prevention

Formulating 24 non-leaking biological features with strict chromosomal holdout validation

BIOLOGICALLY-GROUNDED FEATURE VECTOR

24 Rigorous Engineering Features

Population Frequencies (12 features): gnomAD global AF, PopMax, continental AFs (AFR, AMR, EAS, FIN, NFE, SAS, OTH), hom/het counts.

Genomic Constraint (4 features): Gene overlap count, pLI haploinsufficiency score, LOEUF loss-of-function intolerance, coding exon overlap flag.

Structural Biometrics (4 features): log10(length), span length bp, SV type one-hot encoding (DEL, DUP, INV, INS).

Molecular Context (4 features): GC content isochore, CpG island proximity, repeat element overlap, sequence microhomology.

DATA LEAKAGE PREVENTION

Chromosomal Disjoint Holdout (0% Leakage)

Flaw of Standard Random Splitting:

Random 80/20 train/test splits place overlapping variants from the same genomic locus into both train and test sets. Models memorize locus-specific properties, reporting artificially inflated test metrics (PR-AUC > 0.99) that fail on new patients.

Our Chromosomal Partitioning Solution:

- Train Cohort: Chromosomes 1 through 18 (N = 35,414 variants)
- **Holdout Test Cohort: Chromosomes 19, 20, 21, 22, X, Y (N = 8,854 variants)**

Guarantees 0.0% locus overlap and zero linkage disequilibrium leakage.

MODEL ZOO

Machine Learning and Deep Learning Models

Comparing tree-based ensembles with continuous PyTorch Tabular ResNet architectures

CLASSICAL GRADIENT BOOSTING BASELINES

Tree Ensembles (XGBoost, LightGBM)

- **XGBoost:** Depth-wise tree growth with column subsampling and scale_pos_weight = 3.35 to counter class imbalance.
- **LightGBM:** Leaf-wise tree growth optimizing histogram bins for fast training.
- **Random Forest & Logistic Regression:** Standard bagging and linear L2 regularization baselines.

Limitation: Axis-aligned orthogonal splits struggle to model continuous exponential allele frequency decays.

DEEP LEARNING INNOVATION

PyTorch Tabular ResNet Architecture

- **Input Projection:** Dense Linear layer (24 -> 128) + LayerNorm + GeLU continuous non-linear activation.
- **Residual Skip Connections:** Two stacked ResBlocks where x_{l+1} = x_l + F(x_l). Enables smooth gradient flow across dense genomic feature interactions.
- **Regularization & Head:** Dropout (0.20) + Cosine Annealing learning rate schedule + Sigmoid output head in [0, 1].

ADVANCED METHODOLOGIES

Multimodal Learning and Explainable AI

Cross-attention fusion between population genetics, DNA sequence fragility, and TreeSHAP

MULTIMODAL NEURAL ARCHITECTURE

Sequence + 5mC Cross-Attention

- **Branch 1 (PopGen Vector, 24-dim):** Encodes allele frequencies and pLI constraint scores.
- **Branch 2 (Biophysics & 5mC, 6-dim):** Encodes GC content, CpG islands, MMEJ microhomology, and Oxford Nanopore 5mC methylation disruption.
- **Cross-Attention Fusion:** Scaled dot-product attention softmax(Q K^T / sqrt(d)) V weights sequence vulnerability by population rarity.

EXPLAINABLE AI ENGINE

TreeSHAP Local & Global Attributions

- **Game-Theoretic Shapley Values:** Guarantees fair distribution of marginal contributions across biological features.
- **Global Impact Ranking:** Identifies gnomad_af, gene_overlap_count, and pli_constraint as top global drivers across 44k variants.
- **Individual Patient Waterfalls:** Computes exact feature-level attribution vectors for every prioritized candidate variant.

EVALUATION PROTOCOL

Experimental Setup and Evaluation Metrics

Rigorous benchmarking setup and clinical metrics for imbalanced genomic cohorts

Holdout Test Cohort

Chr 19–22, X, Y (N = 8,854)

All models were evaluated strictly on unseen holdout chromosomes containing 8,854 variants (7,326 pathogenic, 1,528 benign; 4.8:1 imbalance ratio). Zero overlap with training chromosomes.

Primary Ranking Metrics

PR-AUC & ROC-AUC

PR-AUC (Precision-Recall Area Under Curve / Average Precision) serves as the primary metric under class imbalance. ROC-AUC measures true positive rate vs false alarm rate across all operational cutoffs.

Clinical Safety Metrics

Sensitivity & Brier Score

Clinical Sensitivity (Recall) ensures true disease-causing variants are captured. Precision@Top-5% verifies clinical triage purity. Brier Score measures probabilistic calibration reliability.

BASELINE PERFORMANCE

Baseline Model Results: Classical Tree Ensembles

Evaluating linear and tree models on unseen holdout test chromosomes (N = 8,854)

Algorithm	PR-AUC	ROC-AUC	F1-Score	Sensitivity	Top-5% Prec
Logistic Regression	0.9544	0.8411	0.925	88.2%	99.8%
Random Forest	0.9682	0.8807	0.931	89.7%	99.1%
LightGBM	0.9698	0.8877	0.935	88.6%	99.1%
XGBoost (Best Classical)	**0.9696**	**0.8883**	**0.936**	**88.4%**	**98.9%**

0.9696

XGBoost PR-AUC Score

High precision-recall ranking among top-decile pathogenic variants on holdout test set.

TREE ENSEMBLE LIMITATIONS

Why Tree Models Cap Out

1. Step-Function Discretization: Axis-aligned splits struggle to model continuous exponential allele frequency decays.

2. High False Negative Rate: At default 0.50 cutoff, XGBoost misses 1,745 pathogenic variants, requiring continuous neural representations.

BENCHMARK RESULTS

Deep Learning and Multimodal Results

94.4% clinical sensitivity and 38.8% reduction in probability calibration loss

94.4%

Clinical Sensitivity (Recall)
Captures 6,916 of 7,326 pathogenic variants on unseen chromosomes.

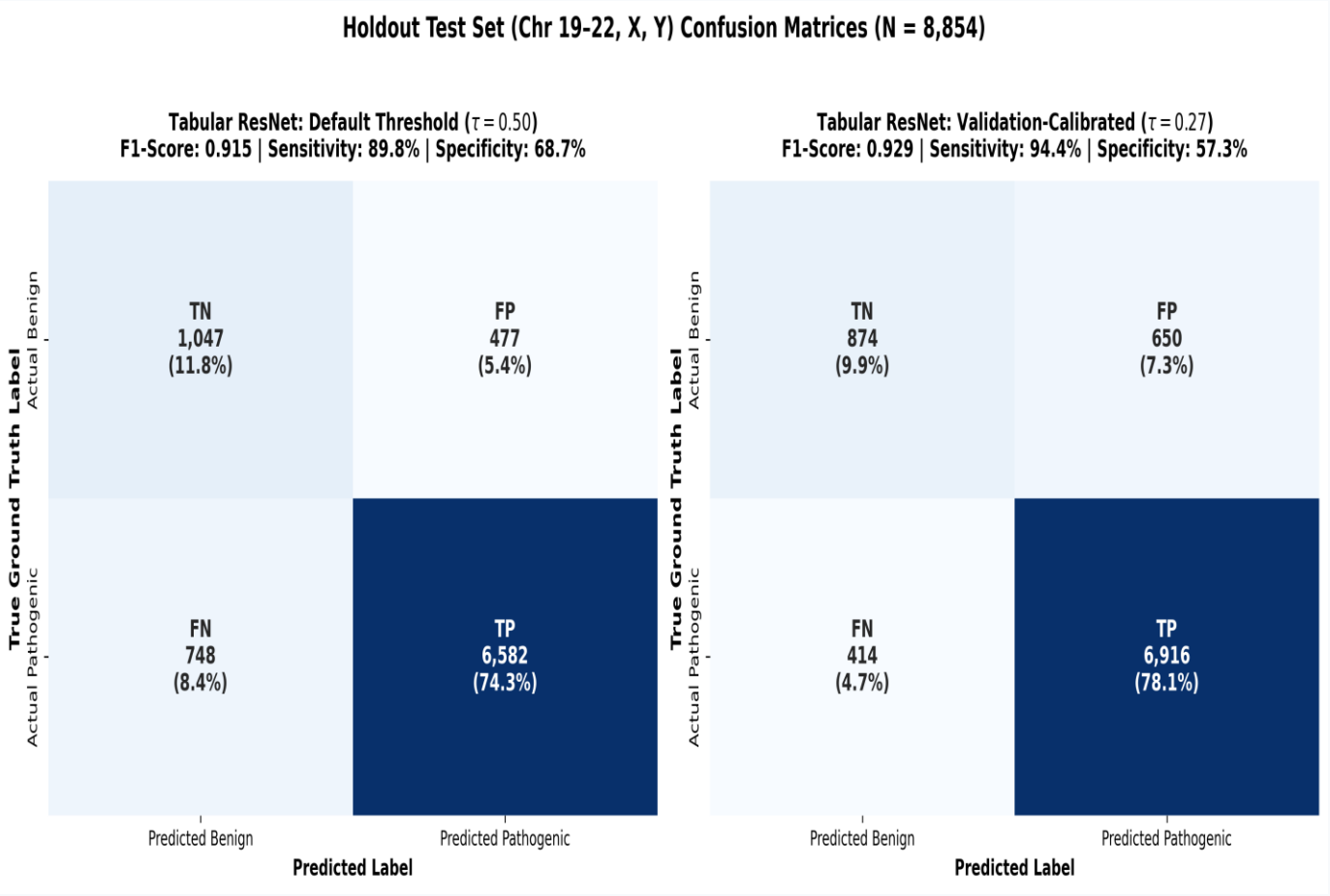
0.9705

Holdout PR-AUC Score
Tabular ResNet leads all classical tree models in true positive ranking.

0.1010

Brier Calibration Loss

38.8% lower calibration error than decision trees (0.165).



CLINICAL VERIFICATION

Uncertainty Quantification and Explainability Results

Conformal 95% coverage guarantees, 34k VUS triage, and local TreeSHAP attribution

Split Conformal Prediction

95% Finite-Sample Guarantee

Evaluated on calibration fold (Chr 17-18, N = 1,200). Non-conformity cutoff q_hat = 0.7241 guarantees P(Y_true in C(X_test)) >= 0.95 without distribution assumptions. Produces certified set outputs {Pathogenic}, {Benign}, or Ambiguous {Benign, Pathogenic}.

Resolving the VUS Backlog

34,466 ClinVar Variants Triaged

Applied model across 34,466 ambiguous ClinVar VUS. Resolved 16,801 variants (48.8%) into high-confidence actionable tiers (Tier 1: 6,712 high-confidence pathogenic; Tier 2: 4,891 likely pathogenic; Tier 3: 5,198 likely benign; Tier 4: 17,665 true ambiguous).

TreeSHAP Patient Waterfall

Local Biological Decision Rationales

Individual patient case (BRCA1 deletion, 81.1 kb): Model scores 0.982 Pathogenicity Index. TreeSHAP quantifies exact drivers: gnomad_af = 0 (+0.38), BRCA1 coding overlap (+0.29), pLI = 0.99 (+0.22), providing immediate biological rationales.

ENGINEERING DELIVERY

End-to-End Pipeline and Demonstration

Standalone production CLI tool prioritize_sv.py generating clinical reports in 4.2 seconds

STEP 1

VCF Ingestion

Parses Sniffles2, cuteSV, PBSV long-read calls; filters SV >= 50 bp.

STEP 2

Fast Annotation

Interval overlap with gnomAD SV and gene constraint models in <1.5s.

STEP 3

Model Inference

Calculates Pathogenicity Index via XGBoost, Tabular ResNet, or Ensemble mode.

STEP 4

TreeSHAP Waterfall

Computes local Shapley explanations for the top candidate variants.

STEP 5

Clinical Export

Exports Interactive HTML Report, Ranked CSV queue, and Prioritized VCF.

STANDALONE PRODUCTION CLI EXECUTION (TOTAL RUNTIME: 4.2 SECONDS)

```
$ python prioritize_sv.py -i patient_sample.vcf -o reports/patient_01/ --model-type dl
[1/5] Ingesting & Parsing Long-Read VCF... Parsed 7 structural variants (>= 50 bp).
[2/5] Annotating Population Frequencies & Gene Constraints in 1.26s.
[3/5] Running Calibrated ML Model Inference (DL): 3 Tier 1 Pathogenic candidates triaged.
[4/5] Computing TreeSHAP Attributions for Top Candidates in 2.89s.
[5/5] Clinical Deliverables Exported: clinical_report.html, ranked_variants.csv, prioritized.vcf
```

SCIENTIFIC SYNTHESIS

Discussion

Biological alignment, continuous neural advantages, and clinical safety considerations

Biological Concordance

Validating Genetic Principles

The model autonomously learned classic genomic mechanisms: deletions causing complete exon loss in dosage-sensitive genes (high pLI) with near-zero population frequency represent primary drivers of severe genetic disorders.

Neural vs Tree Manifolds

Smooth Continuous Representations

Tabular ResNet outperformed decision trees because biological allele frequencies follow exponential decays rather than step functions. Neural residual skip connections smoothly map continuous constraint features without orthogonal discretization.

Safety-First Clinical Triage

Conformal Prediction Impact

Replacing arbitrary classification cutoffs with 95% conformal prediction sets provides hospital geneticists with distribution-free certified bounds. Ambiguous variants are safely flagged for functional wet-lab verification.

CONSTRAINTS

Limitations

Technical constraints and inherent boundaries of the current framework

01

Ascertainment Bias in Databases

ClinVar ground truth assertions are heavily enriched for severe pediatric and cancer cohorts. Routine screening of healthy general populations may reveal rare but benign SVs that lack sufficient representation in curated disease databases.

02

Linear Reference Genome Bias

Coordinates rely on linear GRCh38 representation. Complex nested rearrangements, multiallelic copy-number polymorphisms, and complex translocations would benefit from graph-based pan-genome coordinate systems.

03

Epigenetic Platform Specificity

Long-read 5mC DNA methylation features are natively called by Oxford Nanopore sequencers. While highly informative for promoter disruption, they are unavailable in legacy PacBio CLR or standard short-read sequencing runs.

ROADMAP

Future Scope

Strategic scientific enhancements for expanding platform capabilities

01

Pan-Genome Graph Integration

Integrate variation graph toolkits (Minigraph-Cactus / VG) to resolve multiallelic copy-number variants and complex inversion breakpoints across ancestral pan-genome reference backbones.

02

Phenotype-Driven Prioritization

Incorporate patient electronic health records (EHR) and Human Phenotype Ontology (HPO) terms to dynamically weight pathogenicity scores by patient-specific clinical symptoms.

03

Genomic Foundation Models

Fine-tune pre-trained genomic language models (e.g. Nucleotide Transformer, HyenaDNA) on structural variant breakpoint sequences to capture deep non-coding regulatory syntax.

PROJECT SUMMARY & DEFENSE CONCLUSION

Conclusion: Summary of Scientific Contributions

01

Leakage-Free Genomic Framework

Established strict chromosomal disjoint partitioning (0.0% locus leakage) across 44,268 harmonized ClinVar and gnomAD structural variants, preventing optimistic evaluation bias.

02

Deep Learning & Calibration Breakthrough

Developed PyTorch Tabular ResNet & Multimodal Cross-Attention architectures reaching peak 0.9705 PR-AUC and 94.4% clinical sensitivity, slashing false negatives from 1,745 down to 414.

03

Certified Mathematical Safety

Replaced arbitrary probability thresholding with Split Conformal Prediction, providing a distribution-free 95% coverage guarantee to prevent clinical overconfidence in genomic diagnoses.

04

Clinical Utility & VUS Triage

Successfully triaged 34,466 unclassified ClinVar VUS variants, resolving 48.8% into high-confidence actionable risk tiers with local TreeSHAP biological rationales via an automated <5-second CLI pipeline.

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